

Ann Rose Edakkalathur

AI / Machine Learning Engineer | Generative AI Engineer

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SUMMARY

AI/ML Engineer with around **3 years of experience** in **software development** and hands-on experience in **machine learning** and **generative AI**. Strong background in **Python, Java, NLP**, and building **AI models**, with practical experience from **Infosys** and **IBM**. Skilled in developing, deploying, and improving AI solutions using cloud platforms like Azure and AWS.

PROFESSIONAL EXPERIENCE

AI/ML Intern

Jul 2025 – Dec 2025

IBM – USA

- Worked on building AI agents and supporting frameworks for complex system design and reasoning tasks.
- Customized and fine-tuned NLP models, including BERT for named entity recognition (NER) using PyTorch, improving IOB classification precision by 30% by analyzing incorrect labeling patterns.
- Developed and fine-tuned LLM based AI agents to cross validate BERT predictions, further improving overall model precision.
- Processed and analyzed 40,000+ JSON documents using Python, NumPy, and Pandas to generate insights from job title based datasets and support business decisions.
- Assisted in developing machine learning models for predictive maintenance and production optimization, contributing to improved operational efficiency.
- Implemented model explainability techniques such as SHAP and LIME and evaluated multiple frameworks to better understand model behavior.

System Engineer

Nov 2022 – Sept 2024

Infosys Ltd. – INDIA

- Designed and deployed Java Spring Boot based features and REST APIs for an internal employee attendance portal as part of the Orbit project at Infosys, improving operational efficiency and user experience by 25%.
- Led a smooth Java version upgrade and implemented new features for Medicare software while working with Beacon Healthcare Systems in the USA, resulting in a 30% improvement in data accuracy and overall system performance.
- Engineered backend Java modules using Spring Boot and Spring Security for a large scale social media platform developed for the Professional Forum project under NEOM, Saudi Arabia.
- Delivered solutions using Microsoft Power Platform to build custom websites, automate workflows, and develop mobile apps for The Line project at NEOM, Saudi Arabia, ensuring zero downtime during implementation.
- Worked on fintech applications for a multinational investment banking client using Java Spring Framework and ReactJS, following Agile development practices to deliver reliable features.
- Optimized SQL queries and implemented indexing and stored procedures to improve application performance and reduce data retrieval time.
- Collaborated with cross functional teams, participated in code reviews and Agile ceremonies, and ensured timely delivery of high quality backend features.

TECHNICAL SKILLS

Programming Languages:	Python, Java, SQL, R, HTML, CSS, JavaScript
Machine Learning & AI:	Machine Learning, Deep Learning, Natural Language Processing (NLP), Predictive Modeling, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation
Generative AI & NLP:	Large Language Models (LLMs), Prompt Engineering, Text Classification, Named Entity Recognition (NER), Retrieval Augmented Generation (RAG), Embeddings
Frameworks, APIs & Libraries:	TensorFlow, PyTorch, Scikit Learn, XGBoost, NumPy, Pandas, Hugging Face, LangChain, OpenAI APIs, Spring Boot, REST APIs
Data & Databases:	PostgreSQL, MongoDB, SQL, NoSQL, Hadoop, Data Analysis
Cloud & Deployment:	Microsoft Azure, AWS (EC2, S3, Lambda), GCP Azure AI Services, Azure OpenAI, Model Integration, CI CD, Monitoring and Logging
Tools & Development:	GitHub, Postman, IntelliJ IDEA, VS Code, Eclipse, Android Studio, MATLAB, Tableau, Google Workspace, Microsoft Power Automate, Microsoft Page Automate, MS Office
Operating Systems:	Windows, Linux, Android, iOS

PROJECTS

Drive Smart – Enhancing Vehicle Safety and Efficiency with Predictive Maintenance

- Developed a machine learning model to predict vehicle maintenance needs by evaluating multiple algorithms.
- Deployed the model as a Flask based web application for real time predictions, enabling proactive maintenance scheduling to improve cost efficiency and fleet safety.

Cancer Detection Using AI

- Built a machine learning model to analyze MRI brain scan images and predict the presence of tumors. Collected and pre processed a dataset of brain tumor images and implemented suitable machine learning algorithms.
- Achieved 80% accuracy in predicting tumor presence on new and unseen images.

Home Rentals Application

- Created a full stack web application with secure login using JWT and role based access for renters, property owners, and admins.
- Used MongoDB for efficient data storage and implemented features such as property listings, search and filter, and a real time booking system to improve user experience.

EDUCATION

Masters in Computer science

Mercy University, Dobbs Ferry, New York

Sept 2024 - Dec 2025

CERTIFICATIONS

Microsoft Azure Fundamentals (AZ 900)
Microsoft Azure Administrator Associate (AZ 104)
Microsoft Azure AI Fundamentals (AI 900)
AI for All: From Basics to GenAI Practice – NVIDIA